

Varen 6869 AgSp1.1 cr	PGNNPGVEMGPDGQINKIILPSTPANPSSSGNPNG..Q.....	36
Varen 6869 AgSp1.2 cr	PGNTPGAELGPDGQINRIILPSTPPNPSSSGPLNPDGEP.SAPFGPGNSFNSPK.....SPGFKFPGFPGAPGSMGPGVGMDFSQLPSGESPE	87
Mg TRINITY DN1546 c0 g1 Cr	GGNTPGAELDPDGQIKKVIILPSTPSEPFSSSSSLPSQ.....DSQLNGREPYE	47
Msag 19329 AgSp1.1 cr	PGHTPGAELGPNGNINRIFLPTAPNPSPGPLNPPGEP.IAPFGPGNSPNSP.QSPGNYPGSAFQFPFGFPGAPGSNGPIGYIDFSQLPNGQPPE	93
Msag 18528 AgSp1.2 cr	PGTTPGVVTGSDG.....QPVK.....Y	18
Ncruc 58532and13532 AgSp1 cr	PGTTPGVVRIDPDGKINNVLVLPISPNSELSGPLNPDNEP.IPPFGPDNTPNSP.NFPGNYPRSTFKFPGFPGAPGSMGPIGFLDFSELS..ESPE	91
Lcorn 51676 AgSp1.1Cfrag cr	PGNTPGAELGPDGQINRIILPSTPPNPSSSGPLNPAGEP.IAPFGPGNSPNSP.QSPGNYPGSAFQFPFGFPGAPGSNGPIGYLDFSQPLSDVSPE	95
Lcorn 58650 AgSp1.2 cr	PGNTPGAELGPDGQITRIILPSTPPNPSSSGPLNPAGEP.IAPFGPGNSPNSP.QSPGNYPGSAFQFPFGFPGAPGSNGPIGYLDFSQPLSDVSPE	93
Atri AgSp1 genomic cr	PGTTPGAELGPDGKINKIILPSTPPNPSSSGPLNPKGEP.IPPFGPGNSPNSP.QPPGNYPGSAFQFPFGPDAPGANGPIGYIDFSQLPSGESPE	93
Amarm 50219 AgSp1.1 cr	PGNTPGAQLGPDGQINRIILPSTPPNPSSSGPLNPAGEP.TAPFGPGNSPNSP.QSPGNYPGSAFKFPGFPGAPGANGPIGYIDFSQLPSGESPE	93
Aarg AgSp1 FC511 cr	PGTTPGAELGPDGKINKIIVLPTTPNPSSSGPLNPEP.IPPFGPGNSPNSP.QTPGNYPGSAFQFPFGPDAPGSNGPIGYIDFSQLPSGESPE	93
Mhut 1271 AgSp1 cr	PGITPGVELGPDGQINRIILPSTPPNPVLFSDPLNPVKGK.AAPFGPGNIPNSP.YSPGNYPGSAFQFPFGFPGAPGSNGPIGYIDFKQLPSGVSPE	93

Varen 6869 AgSp1.1 cr	SFRPSGCDPSNLGPDGSLNTDSIPDFDNPGYGSFGYPQVPLGYFDSVP.....NSDPSSQFVLPGSPGSPGSDGQFPFG	109
Varen 6869 AgSp1.2 cr	MYGSIGFLPNFSPEIGGPFPGFPGGPGDFGPGGYLNTDSLPDFINPGYGFPGSPQAPLGDFDNKLPDFNSDFSGMFVLPGSPGSPGSGNGQFPFG	182
Mg TRINITY DN1546 c0 g1 Cr	MDDSVGHFPNFNLEIGGSFLGYEGGPNNLGLGENLNIEITLPYFIKPGSGFPGSSHVFPFGHFDYNQLPPDFNPNFPGLFALPGYSGFPESSRSQFAG	142
Msag 19329 AgSp1.1 cr	QVGSIGYFPDFSGEIGGPFPGFSGGPGGPGGGYLNIMSLPDFTNRGYGFPGSPQAPLGYLDESMLPKGFNPNFSGGFVYPGYPGSPGSGGGQYYPG	188
Msag 18528 AgSp1.2 cr	IAPPGGISPNFGLGIEIGGVFSPISAGADDSAPGEHLNIHSIPNLINPLYGFPSFQAPLGIFYSSQ.....SSEFSDKFVLPGDLLSPGSEGGFLG	108
Ncruc 58532and13532 AgSp1 cr	FDGSIGFFPNFNPEIGGPFPGFSGGPGEDSGPGGYLDIDSIPDFINPLYGFPGSPQAPLGYIHNHP.....SSDVSDTFVLPGDPGAPGSGQGFPG	181
Lcorn 51676 AgSp1.1Cfrag cr	MVGNIGFFPNFNFSPEIGGPFPGFPGGPDDSGPGGFLNIGSLPNFKSPGYGFPGSPQAPLGYLQNMPLPGYNPDFSGPLVFPGYSGSPGSGGQLPG	190
Lcorn 58650 AgSp1.2 cr	MVGPIGFFPNFNPEIGGQFPFGPGGPDDSGPGGFLNINSLPDFRNPYGFPGSPQAPLGYLDESMLPQDFDFDFSGGFVFPGYPGSPGSGSQFPFG	188
Atri AgSp1 genomic cr	MVGPIGFFPNFNPEIGGQFPFGPGGPDDSGPGGFLNINSLPDFMNPYGFPGSPQAPLGYLDESMLPQDFSLDFSGGFVFPGYPGSPGSGSQFPFG	188
Amarm 50219 AgSp1.1 cr	MEGPIGFLPNFSPEIGGPFPGFPGGPGSSGPGGFLNVNSLPDFINQYGFPGSPQDPLGYLNFMSMLPPGYSPDFSGGFVYPGYPGSPGSGGGQFPFG	188
Aarg AgSp1 FC511 cr	MVGSIGFFPNFNFSPEIGGPFPGFSGGPGNSGPGGFLNINSLPDFMNPVYGFPGSPQAPLGYLKSMPLPPGFNLDSGGQLVFPDGPSPGSGSQFPFG	188
Mhut 1271 AgSp1 cr	MEGSIGFLPNFSPEIGGPFPGFPGGPGSPGLSGFLNVNSLPDFINQYGFPGSPQDPLGYLNFMSMLPPGYNPDFSGGFVYPGYPGSPGSGGGQFPFG	188
	MIGSIGFLPNFSPEIGGPFPGFPGGPDDSGPGGYLNIDSLPDFNKPQYGFPGSPQAPLGYLKSKLPPGYNPGFSGQLVFPGYPGAPGSGSQFPFG	188

Varen 6869 AgSp1.1 cr	GFLSFMQLPKSITDKINRTFDFTELIQAVQPLPFGNTINLGAIPRDNLQDIPGFDGSYNNLKFDYFGDSTKQNGGIFYIPELIYLINQLPG....	200
Varen 6869 AgSp1.2 cr	GFLSFPQLPKDITDKIDRPFSFPQLIQAIQPLPFGDSINVGAIPEDKLDKMPGFNGTSDSLRLDNFEDNNRPSGGVFYIPELVRLISYLPVGSFG	277
Mg TRINITY DN1546 c0 g1 Cr	GFLSFPQLPKGITNNINGSFSFAELIQALQPLPFGKSINVGAIPQNKLVMPGFNGTYDNLGLDIDGSSHPNGGVFYLPVLRLVGYLPVGSFG	237
Msag 19329 AgSp1.1 cr	GFLSFSQLPESITNIHGSFSFSQMMNALGPLPFGQ.VNIGAIPKNQLQSI PGFNGNYDNLMLDNYGDSSQPTGGVFYLPVLRLIASYLPESFS	282
Msag 18528 AgSp1.2 cr	GFLSFPQLSKDITNLINGSFSFAELIQALQPLPFGNSINVGAIPHSELEVMPGFNGTYDNLKLDNIKNNGHPNGGVFYLPVLRLIDFVVPVGSFN	203
Ncruc 58532and13532 AgSp1 cr	GFLSFPQLPKSITDMINGTFSFQELIQALQPLPFGNSINVGAIPDQKLQDIPGLKGNHDSLKLSQLGDNSQPSGSLFYIPELMILISHLPD....	272
Lcorn 51676 AgSp1.1Cfrag cr	GFLSFPQLPKDITDKISSTFSFPELIKSLQPLPFGQINMGAIPQDKLQNI PGFNGTYDNLKLANYGDNQPTGGVFYLPVLRLMSYLPVGSFG	285
Lcorn 58650 AgSp1.2 cr	GFLSFPQLPKDITDKINGTFSFPQLINALQPLPFGQINIGAIPQDKLQNI PGFNGSYDNLKLANFGDNSQPTGGVFYLPVLRLISYLPVGSFG	283
Atri AgSp1 genomic cr	GFLSFSSELPQGVTDKINGTFSLPQLMQALQPLPFGQINIGAIPKQMQNI PGFKGTYDNLRLANFGSDNQPTGGVFYLPVLRLTSYLPVGSFG	283
Amarm 50219 AgSp1.1 cr	GFLSFPQLPKIEITDKIDGTFSFPQLIEALQPLPFGQINIGAIPKDKLQNI PGFEGTYDNLKLANFGGNNQPTGGVFYLPVEMVRLIGYLPVGSFG	283
Aarg AgSp1 FC511 cr	GFMSPPELPKDVTDKINGTFSLPQLMQALQPLPFGQINIGAIPKQMQNI PGFEGTYDNLRLANFGNNNQPTGGVFYLPVLRLISYLPVGSFG	283
Mhut 1271 AgSp1 cr	GFLSFPQLPKDITDKISGTFSFPQLINSLQPLPFGQINIGTIPQDKLQNI PGFTGTYNLKLKLANFGDNSQPTGGVFYLPVLRLIGYLPVSSS	283

Varen 6869 AgSp1.1 cr	GISGT F DFPGFYGPRRN I CDAG N ...D S SDSGSQDLGGEVE G DL D GLFGD M GDAAPSDT M DAAPR.....	263
Varen 6869 AgSp1.2 cr	K GS G TINQDGG F CE P FN F PGLNGAP G N I CDY P EDY D Y S MD E GSQDLGGEVE G D G NGPSGD V CD A APED I GA A P P NNLSP.....	358
Mg TRINITY DN1546 c0 g1 Cr	I GP G TIN Q EGGF G DA I K F PGLNGAPGY I CDY P ENG... D ATDRGSQDLGEG V FG D GNTD T ND D GF Q ES.....	302
Msag 19329 AgSp1.1 cr	 A NGGF G DP F GE Q LL H GAPGY I CDY P DND... D TT A GG S QDLG G E T K G NG H G P SGDV A DA A P... N L G G S AP G NDL G GA A P GNT	359
Msag 18528 AgSp1.2 cr	K GL G TIN Q EGGF G HP V N F PGL S GT L GY I CDY L ENG... D AT I EG S Q I FS.....	249
Ncruc 58532and13532 AgSp1 cr	I GP E TIN S DGG F ST P FN F P G FS G PPGY I CDY P DKD... D VT D GG F HD L GG G V K GN D NR P FG D V G GA A P D EDD A ND V AS Q DL G GE V K...INK T GP F RD	364
Lcorn 51676 AgSp1.1Cfrag cr	K GP G TIN H DGG F D D PF N F P GLNGAP G S I CDY P DNG... D AT D GG S QDLGGE V K G D G NG P A GD V ADA A P... D L G A P A P G NDL G GA A P D L G A P AGND	376
Lcorn 58650 AgSp1.2 cr	K GS G TINQDGG F GDP F N F PGLNGAPGY I CDY P DNG... D ATAGGSQDLGGE V K G NGNG P SGDV A DA A P... D L G SP A P E NDL G GA A P.....	364
Atri AgSp1 genomic cr	N GP G TIN Q NGGF G DP F N F PGLNGAPGY I CDY P DNG... H AT T GG S Q N L G GE T K G N D NG P SGDV A DA A P G NN L G A AP G NN L G S A A P GN.	368
Amarm 50219 AgSp1.1 cr	Y GP G TIN H DGG F G H PF N F P GLNGAPGY I CDY P N NG... V L... S QDLGGE V K G NGNG P SGDV A DA A P E NDL G G T A P END I G G A A P EN.	364
Aarg AgSp1 FC511 cr	N GP G TINQDGG F GDP F N F PGLNGAPGY I CDY P DNG... D AAT G RS Q N L GGE T K G NGNG P SGDV A DA A PE A DL G A A P Q SD L A.....	362
Mhut 1271 AgSp1 cr	K DSE I IN K NG F S N PF N F P GLNG F PGY I CDY P DNG... D AT T EG S QDLGGE I EGY G NG P FED V ADA A P E NDL E	353

Varen 6869 AgSp1.1 cr		263
Varen 6869 AgSp1.2 cr	P	359
Mg TRINITY DN1546 c0 g1 Cr	GKEVKDNRNGSSGDVGNSFPDYS I V G VT P L T N Q S....SGA	339
Msag 19329 AgSp1.1 cr	 L GGS A P G ..N..... T L G GS A P.....GNAP G CSAPGN N M..A S NT L V G	394
Msag 18528 AgSp1.2 cr	VGGAAPNEDGANDGGSQDLGGEV.KSNK V G P FRD V GGA A PYEDGANDGDSQDLGGE V K G SGNG P SGDV A DA A PDD N MM A D A P Q S N L..SS P T P V T	456
Ncruc 58532and13532 AgSp1 cr	LGGAAPD L GAP A P G ..NDLGGAAPDLGAP A P G ND L GGAAPDLGAP A P..GNDLGGAAADLG.....APAPAPGND L GATDPP S D Q ..GSNT L V A	459
Lcorn 51676 AgSp1.1Cfrag cr	DLG S PA.....P E NDLGG V AP.....NLGSPAP Q T N Q G ..SNT L V A	398
Lcorn 58650 AgSp1.2 cr	DLG S PA.....P E NDLGG V AP.....NLGSPASE T N Q G..SNT L V A	398
Atri AgSp1 genomic cr	NLG S AA.....PGNN L G S AAP.....END L GGAAP E NN P A P Q S D T P S S	406
Amarm 50219 AgSp1.1 cr	DLG G AA.....P E NDLGG A AP.....END L GGAAP Q SE M G..SNT L V A	400
Aarg AgSp1 FC511 cr	S P T P V A	368
Mhut 1271 AgSp1 cr	 G VAS Q S N L G ..P N T L V A	368

Varen 6869 AgSp1.1 cr	.NDP S CG D DD V NS A F S T A K Q AL Q	284
Varen 6869 AgSp1.2 cr	TSV T LC E DN V ST F SV A K Q AL Q	381
Mg TRINITY DN1546 c0 g1 Cr	DI D GN C D N D V ATT F NT T K Q A I R	361
Msag 19329 AgSp1.1 cr	P E D EG C QDN V PE A FS K AR Q AL A	416
Msag 18528 AgSp1.2 cr	GT E GN C ED D VD R TL N T A K Q AL Q	478
Ncruc 58532and13532 AgSp1 cr	P DE D EDCQDN V PE A F H KAR Q AL A	481
Lcorn 51676 AgSp1.1Cfrag cr	P DE D EDCQDN V P E T F SKAR E AL A	420
Lcorn 58650 AgSp1.2 cr	P D N EDCQDN V P E T F SKAR E AL A	420
Atri AgSp1 genomic cr	P T D K G CE E D V Q G A F SKAR Q AL M	428
Amarm 50219 AgSp1.1 cr	P E D EDCQDN V PE A F N KAR Q A T	422
Aarg AgSp1 FC511 cr	P T D Q D CE E D V H G A F SKAR Q AL M	390
Mhut 1271 AgSp1 cr	P E V K D CQDS V PE A FR K AR H ALT	390

☒ non-conserved
☒ similar
☒ ≥ 50% conserved